

Discrete Ordinates Method for Leaf-Canopy Radiative Transfer

A Self-Study Guide: Quadrature, Diamond Differences,
Source Iteration, and Extensible Design

EE645 – Physical Models in Remote Sensing

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1 The Radiative Transfer Equation

The radiative transfer equation (RTE) describes how the specific intensity $I(\mathbf{r}, \boldsymbol{\Omega})$ [$\text{W m}^{-2} \text{sr}^{-1}$] changes along a ray traveling in direction $\boldsymbol{\Omega}$:

$$\frac{dI}{ds} = -\kappa I + j \quad (1)$$

where s is path length, κ is the extinction coefficient, and j is the emission/scattering source. For a *plane-parallel leaf canopy* parameterized by the cumulative leaf area index L (0 at canopy top, L_{AI} at ground), the equation becomes

$$\mu \frac{\partial I(L, \boldsymbol{\Omega})}{\partial L} = -G I(L, \boldsymbol{\Omega}) + \frac{G \omega_L}{\pi} \int_{4\pi} \Gamma(\boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}) I(L, \boldsymbol{\Omega}') d\boldsymbol{\Omega}' \quad (2)$$

where:

- $\mu = \cos \theta$ is the polar cosine of direction $\boldsymbol{\Omega}$ ($\mu < 0$ downward, $\mu > 0$ upward)
- $G = 0.5$ for a uniform (spherical) leaf-angle distribution
- $\omega_L = \rho_L + \tau_L$ is the leaf single-scattering albedo
- $\Gamma(\boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega})$ is the bi-Lambertian scattering phase function [sr^{-1}]

The integral over all directions couples every direction to every other direction. This is the fundamental difficulty – it cannot be solved analytically in the general case.

1.1 Bi-Lambertian Phase Function

The scattering angle β between incident direction (μ_i, ϕ_i) and scattered direction (μ_j, ϕ_j) satisfies:

$$\cos \beta = \mu_i \mu_j + \sqrt{1 - \mu_i^2} \sqrt{1 - \mu_j^2} \cos(\phi_i - \phi_j) \quad (3)$$

The bi-Lambertian phase function (Myneni et al.) is:

$$\Gamma(\boldsymbol{\Omega}_i \rightarrow \boldsymbol{\Omega}_j) = \frac{\omega_L}{3\pi} (\sin \beta - \beta \cos \beta) + \frac{\tau_L}{3} \cos \beta \quad (4)$$

The first term is diffuse (Lambertian) reflection; the second is forward transmission through leaves.

2 Gauss–Legendre Quadrature

2.1 The Core Idea

We replace the continuous directional integral in (2) with a finite weighted sum:

$$\int_{-1}^1 f(\mu) d\mu \approx \sum_{i=1}^N w_i f(\mu_i) \quad (5)$$

The question is: *where do we place the N nodes μ_i , and what weights w_i do we assign?*

2.2 Why Legendre Polynomial Zeros?

Fundamental theorem of Gaussian quadrature: Among all possible choices of N nodes and N weights, the choice that maximizes the polynomial degree integrated exactly is achieved if and only if the nodes $\mu_1, \dots, \mu_N$ are the zeros of the N -th Legendre polynomial $P_N(\mu)$.

With this choice the rule is *exact* for all polynomials of degree $\leq 2N - 1$, using only N function evaluations. Uniform spacing can only achieve exactness up to degree $N - 1$ (the trapezoidal rule).

2.3 The Legendre Polynomials

The Legendre polynomials $P_n(\mu)$ are orthogonal on $[-1, 1]$:

$$\int_{-1}^1 P_m(\mu) P_n(\mu) d\mu = \frac{2}{2n+1} \delta_{mn} \quad (6)$$

They are generated by the three-term recurrence:

$$P_0(\mu) = 1 \quad (7)$$

$$P_1(\mu) = \mu \quad (8)$$

$$(n+1) P_{n+1}(\mu) = (2n+1) \mu P_n(\mu) - n P_{n-1}(\mu) \quad (9)$$

The first few roots (for $N = 4$) are approximately $\mu \in \{-0.861, -0.340, +0.340, +0.861\}$, which are symmetric about zero and cluster toward the endpoints – denser sampling near horizontal and near-zenith directions.

2.4 Computing the Weights

Given a node μ_i (a zero of P_N), the corresponding weight is:

$$w_i = \frac{2}{(1 - \mu_i^2) [P'_N(\mu_i)]^2} \quad (10)$$

NumPy computes these via the *Golub–Welsch algorithm*: it constructs the symmetric tridiagonal matrix whose eigenvalues are the Gauss nodes and whose eigenvectors give the weights. This is numerically stable for large N .

```
1 # numpy's built-in: returns nodes in ascending order on (-1, 1)
2 nodes, weights = np.polynomial.legendre.leggauss(N)
```

2.5 Azimuthal Quadrature

For the azimuthal integral over $\phi \in [0, 2\pi)$, we use M *uniform* points because the integrand is periodic and smooth:

$$\phi_m = \frac{(m-1)2\pi}{M}, \quad w_\phi = \frac{2\pi}{M}, \quad m = 1, \dots, M \quad (11)$$

This is the rectangle rule, which converges exponentially for smooth periodic functions.

2.6 Index Convention (1-Based)

The code uses 1-based arrays with a dummy index-0 slot so that array index k maps directly to the notation $k = 1, \dots, N$:

```

1 mu      = np.zeros(N + 1)    # index 0 unused
2 mu[1:N+1] = nodes           # nodes in ascending order
3 # mu[1..N/2] < 0   (downward)
4 # mu[N/2+1..N] > 0   (upward)

```

3 The Uncollided / Collided Decomposition

3.1 Motivation

The direct solar beam is a *delta function* in angle – infinitely concentrated in one direction. Representing it on a finite quadrature grid would require enormous N for accuracy. Instead we isolate it analytically:

$$I(L, \Omega) = I_0(L, \Omega) + I_c(L, \Omega) \quad (12)$$

I_0 : Uncollided field. Radiation that has never scattered. No source term – just exponential attenuation. Solved analytically.

I_c : Collided field. Radiation that has scattered at least once. Smooth in angle – well-represented on the quadrature grid. Driven by the *First Collision Source* Q .

3.2 Uncollided Field: Beer–Lambert

For the direct solar beam (intensity $I_{\text{beam}} = f_{\text{dir}} F_{\text{in}} / |\mu_\odot|$):

$$I_0^{\text{dir}}(L) = I_{\text{beam}} \exp\left(-\frac{G L}{|\mu_\odot|}\right) \quad (13)$$

For isotropic diffuse sky (intensity $I_{\text{sky}} = (1 - f_{\text{dir}}) F_{\text{in}} / \pi$) along each downward Gauss direction (μ_n, ϕ_m) :

$$I_0(L, \mu_n, \phi_m) = I_{\text{sky}} \exp\left(-\frac{G L}{|\mu_n|}\right), \quad n = 1, \dots, N/2 \quad (14)$$

The upward uncollided field comes from Lambertian ground reflection of the total downward flux reaching the ground:

$$I_0^{\text{gnd}} = \frac{\rho_g}{\pi} F_\downarrow(L_{\text{AI}}) \quad (15)$$

and then attenuates upward through the canopy for each upward Gauss direction.

3.3 First Collision Source

When an uncollided photon scatters for the first time it feeds the collided field. The source has three contributions:

$$Q_1(L, \Omega) = \frac{1}{\pi} \Gamma(\Omega_{\odot} \rightarrow \Omega) I_0^{\text{dir}}(L) \quad (\text{direct solar}) \quad (16)$$

$$Q_2(L, \Omega) = \frac{1}{\pi} \sum_{n=1}^{N/2} \sum_{m=1}^M w_n w_{\phi} \Gamma(\Omega_{nm} \rightarrow \Omega) I_0(L, \Omega_{nm}) \quad (\text{diffuse sky}) \quad (17)$$

$$Q_3(L, \Omega) = \frac{1}{\pi} \sum_{n=N/2+1}^N \sum_{m=1}^M w_n w_{\phi} \Gamma(\Omega_{nm} \rightarrow \Omega) I_0(L, \Omega_{nm}) \quad (\text{ground-reflected}) \quad (18)$$

$$Q = Q_1 + Q_2 + Q_3 \quad (19)$$

4 Diamond Difference: Discretizing the Sweep

4.1 Setting Up the 1-D ODE

After fixing a discrete direction (μ_n, ϕ_m) , the RTE reduces to a 1-D first-order ODE in L . We divide the canopy into K uniform layers of thickness $\Delta L = L_{\text{AI}}/K$. The unknown is the intensity at each of the $K + 1$ layer interfaces (edges):

$$\mu_n \frac{I_{k+1} - I_k}{\Delta L} = -G \underbrace{\frac{I_k + I_{k+1}}{2}}_{\text{cell-centre}} + J_{k+1/2} \quad (20)$$

The left side is a finite-difference derivative. The right side evaluates the source J at the cell centre. The **Diamond Difference** approximation is the choice:

$$\boxed{I_{k+1/2} = \frac{I_k + I_{k+1}}{2}} \quad (21)$$

This is the “diamond” – on a mesh plot of space vs. angle, the four points $(I_k, I_{k+1}, I_{k+1/2}, J_{k+1/2})$ form a diamond shape.

4.2 Deriving the Sweep Equations

Define the dimensionless optical half-thickness:

$$f_n = \frac{G \Delta L}{\mu_n} \quad (\text{signed: negative for downward}) \quad (22)$$

Downward directions ($\mu_n < 0$, sweep from $k = 1$ to $k = K$): Rearranging the DD equation to solve for I_{k+1} given I_k :

$$\boxed{I_{k+1} = a_n I_k - b_n J_{k+1/2}} \quad a_n = \frac{1 + (1 - \alpha)f_n}{1 - \alpha f_n}, \quad b_n = \frac{f_n}{G(1 - \alpha f_n)} \quad (23)$$

Since $\mu_n < 0 \Rightarrow f_n < 0$, we get $a_n < 1$ (attenuation) and $-b_n > 0$ (source adds to intensity). Both are physically correct.

Upward directions ($\mu_n > 0$, sweep from $k = K$ to $k = 1$):

$$\boxed{I_k = c_n I_{k+1} + d_n J_{k+1/2}} \quad c_n = \frac{1 - (1 - \alpha)f_n}{1 + \alpha f_n}, \quad d_n = \frac{f_n}{G(1 + \alpha f_n)} \quad (24)$$

Since $\mu_n > 0 \Rightarrow f_n > 0$, we get $c_n < 1$ and $d_n > 0$.

4.3 The Role of α

The parameter $\alpha \in [0, 1]$ controls how the cell-centre value is weighted between the two edges:

$$I_{k+1/2} = (1 - \alpha) I_k + \alpha I_{k+1} \quad (25)$$

α	Name	Properties
0.5	Diamond Difference	2nd-order accurate; may produce negative I
1.0	Step (upwind)	1st-order accurate; always positive

4.4 Sign Convention: Why Signed μ Matters

Using the *signed* cosine in $f_n = G\Delta L/\mu_n$ (not $|\mu_n|$) is critical. For a downward direction $\mu_n = -0.5$:

$$\begin{aligned} f_n &= G \Delta L / (-0.5) < 0 \\ a_n &= \frac{1 - 0.5|f_n|}{1 + 0.5|f_n|} < 1 \quad \checkmark \text{ (attenuating)} \\ \text{source term} &= -b_n J > 0 \quad \checkmark \text{ (source adds intensity)} \end{aligned}$$

If instead $|\mu_n|$ were used, $f_n > 0$ for downward, giving $a_n > 1$ (amplifying – unphysical) and the source term would *subtract* from I (wrong sign). This is a common implementation bug.

4.5 The Fix-Up

Standard DD can produce negative intensities in optically thick cells (large $G \Delta L / |\mu_n|$). Negative intensity is unphysical, so we clip:

```
1 I[k+1] = max(a * I[k] - b * J[k], 0.0)
```

The proper solution is to use more layers (larger K) so that each cell is optically thin. As $\Delta L \rightarrow 0$, DD becomes exact.

5 Source Iteration

The collided field equation contains a self-scattering term – I_c depends on itself through the multiple-scattering source S :

$$S(L, \Omega) = \frac{1}{\pi} \sum_{n,m} w_n w_\phi \Gamma(\Omega_{nm} \rightarrow \Omega) I_c(L, \Omega_{nm}) \quad (26)$$

We solve this by **Source Iteration** (also called Λ -iteration):

1. Initialize: $I_c = 0, S = 0$
2. Compute total source: $J = Q + S$
3. Sweep all $N \times M$ directions using DD to obtain new I_c
4. Compute new S from I_c
5. Check convergence:

$$\epsilon = \max_k \frac{|SI_k^{\text{new}} - SI_k^{\text{old}}|}{\max(|SI_k^{\text{new}}|, \epsilon_0)} < \text{TOL} \quad (27)$$

where $SI_k = \sum_{n,m} w_n w_\phi I_{c,k+1/2}^{n,m}$ is the scalar irradiance (integral of intensity over all directions, no cosine weight)

6. Repeat from step 2

5.1 Convergence Rate

Source iteration converges because each scatter removes energy ($\omega_L < 1$). The spectral radius of the iteration is approximately ω_L :

Band	ω_L	Typical iterations
RED ($\rho_L = 0.06, \tau_L = 0.04$)	0.10	3–5
NIR ($\rho_L = 0.525, \tau_L = 0.450$)	0.975	100–500

For NIR, the slow convergence is well-known and can be accelerated with *Diffusion Synthetic Acceleration (DSA)* – beyond the scope of this course.

5.2 Ground Boundary Condition (Collided)

The downward sweep is completed first. The resulting collided downward flux at the ground is immediately reflected Lambertianly to set the lower boundary condition for the upward sweep *in the same iteration*:

$$I_c^{\text{gnd}} = \frac{\rho_g}{\pi} \sum_{n=1}^{N/2} \sum_{m=1}^M w_n w_\phi |\mu_n| I_c(\mu_n, \phi_m, L_{\text{AI}}) \quad (28)$$

Using the current iteration's downward IC (rather than the previous iteration's) for the ground BC significantly accelerates convergence.

6 Energy Balance

At convergence, the total irradiances are:

$$F_{\downarrow}^{\text{tot}}(L) = F_{\downarrow}^{\text{dir}}(L) + F_{\downarrow}^{\text{dif}}(L) + F_{\downarrow}^{\text{col}}(L) \quad (29)$$

$$F_{\uparrow}^{\text{tot}}(L) = F_{\uparrow}^{\text{unc}}(L) + F_{\uparrow}^{\text{col}}(L) \quad (30)$$

The correct energy balance is:

$$\boxed{F_{\text{in}} = F_{\uparrow}^{\text{tot}}(0) + A_{\text{leaves}} + (1 - \rho_g) F_{\downarrow}^{\text{tot}}(L_{\text{AI}})} \quad (31)$$

where the leaf absorption is:

$$A_{\text{leaves}} = G(1 - \omega_L) \Delta L \sum_{k=1}^K SI_{k+1/2} \quad (32)$$

and $SI_{k+1/2}$ is the scalar irradiance (intensity integrated over all directions without a cosine weight):

$$SI_{k+1/2} = I_{0\ k+1/2}^{\text{dir}} + \sum_{n,m} w_n w_\phi [I_{0,k+1/2}^{n,m} + I_{c,k+1/2}^{n,m}] \quad (33)$$

Note: The balance uses $(1 - \rho_g) F_{\downarrow}^{\text{tot}}$ at the ground (ground-absorbed flux), not the full $F_{\downarrow}^{\text{tot}}$. The ground-reflected portion $\rho_g F_{\downarrow}^{\text{tot}}$ is already accounted for in $F_{\uparrow}^{\text{tot}}$ and A_{leaves} .

7 Alternative Methods and Extensible Design

7.1 Alternative Quadrature Schemes

Double Gauss (recommended upgrade): Apply separate Gauss–Legendre quadrature on each hemisphere $[-1, 0]$ and $[0, 1]$ independently, then combine. Each half-hemisphere integral of degree $\leq 2(N/2) - 1$ is exact. This gives significantly better hemisphere flux accuracy for the same N , and is used in production codes such as DISORT.

Implementation: Apply `leggauss(N/2)` on $[0, 1]$ to get upward nodes, then mirror to get downward nodes.

Gauss–Lobatto: Nodes include the endpoints ± 1 (nadir and zenith). Useful when boundary behaviour dominates, but sacrifices one order of accuracy.

Uniform (equal-weight): Evenly spaced $\mu_i = -1 + (2i - 1)/N$ with equal weights $w_i = 2/N$. Simple but only 1st-order accurate. Use only for debugging.

7.2 Alternative Sweep Schemes

Scheme	Accuracy	Always positive?	Notes
Diamond Difference ($\alpha = 0.5$)	2nd order	No	Standard; needs fix-up
Step / Upwind ($\alpha = 1.0$)	1st order	Yes	Same code, change α
Linear Discontinuous (LD)	2nd order	Yes	$2 \times$ unknowns per cell

Linear Discontinuous Galerkin (LD): Each cell has *two* unknowns – a left-edge and a right-edge value – related by a linear interpolation. The scheme is always positive and 2nd-order accurate. The tradeoff is that the inner loop must solve a 2×2 system per cell rather than a scalar recurrence.

7.3 Suggested Modular Design

When rewriting from scratch, separate concerns into distinct modules:

```
DOM_v2/
  config.py      <- all inputs + method flags (QUADRATURE, SWEEP,
                  ALPHA_DD)
  quadrature.py  <- get_polar_quadrature(N, method)
  phase.py       <- gamma_scalar(), precompute_G_qq(),
                  precompute_G_sol()
  uncollided.py  <- solve_uncollided(), compute_Q()
  collided.py    <- solve_collided() with pluggable sweep_coeffs()
  energy.py      <- scalar_irradiance(), energy_balance()
  brf.py         <- brf_at_view() via DOM interpolation
  main.py        <- orchestrates everything, prints diagnostics
```

Key design principle: `config.py` holds *all* parameters. Every function receives only what it needs. This makes it easy to swap quadrature method or sweep scheme without touching solver logic.

8 Quick Reference: Key Equations

Quantity	Formula
Legendre zeros (nodes)	zeros of $P_N(\mu)$; computed via <code>leggauss(N)</code>
Gauss weights	$w_i = 2 / [(1 - \mu_i^2)(P'_N(\mu_i))^2]$
Azimuthal weight	$w_\phi = 2\pi/M$
Uncollided (direct)	$I_0^{\text{dir}}(L) = I_{\text{beam}} e^{-GL/ \mu_\odot }$
Uncollided (diffuse)	$I_0(L, \mu_n) = I_{\text{sky}} e^{-GL/ \mu_n }$
DD coefficients	$f = G\Delta L/\mu$; $a = (1 + (1-\alpha)f)/(1 - \alpha f)$; $b = f/[G(1 - \alpha f)]$
DD sweep (down)	$I_{k+1} = a I_k - b J_{k+1/2}$
DD sweep (up)	$I_k = c I_{k+1} + d J_{k+1/2}$
Multiple-scatter source	$S_{ij} = \pi^{-1} \sum_{nm} w_n w_\phi \Gamma_{nm \rightarrow ij} I_{c,nm}$
Scalar irradiance	$SI_k = I_0^{\text{dir}}{}_k + \sum_{nm} w_n w_\phi (I_{0,nm,k} + I_{c,nm,k})$
Leaf absorption	$A = G(1 - \omega_L)\Delta L \sum_k SI_k$
Energy balance	$F_{\text{in}} = F_{\uparrow}(0) + A + (1 - \rho_g)F_{\downarrow}(L_{\text{AI}})$